



BIO 004 · HUMAN ANATOMY · SOLANO COLLEGE

Repair Round.

In-class activity · 23 minutes · Pairs

SECTION 1

How Repair Round Runs

Facilitation guide. Five phases, what to say at each, and what to watch for while you circulate.

SECTION 2

Student Capture Sheet

One page, one per student. The only thing that leaves the room.

SECTION 3

Prompt Cards

Four sets, one per Module 1 topic. Cut apart, one card per pair. Swap this section to run the activity on any other unit.

SECTION 1

How Repair Round Runs

FACILITATION GUIDE

Twenty-three minutes. Students arrive having done the pre-work. This activity makes them find out what did not stick, name why, and build something that fixes it. The part they will not do alone is the part a partner forces.

- WHY IT WORKS**
1. Retrieval, not review. Nothing is open until phase three, so the first five minutes are pure recall.
 2. A partner cannot be fooled. Students overrate what they know from rereading. Another person looking at a blank space cannot be talked out of it.
 3. Naming the failure type changes the repair. Not recalling and misapplying need different fixes, and students default to the same fix for both.
 4. A tool has to survive a test. If a partner covers it and you cannot produce the item, the tool is not finished.

Before class

- Print one **capture sheet** per student, Section 2.
- Print and cut one **prompt card set** per pair, Section 3. Use the set matching the topic you are on.
- Blank paper for the redraw. Not their pre-work packet.
- Two pen colours per student. The second colour is load-bearing, it makes corrections visible at a glance.

The five phases

TIMING, GROUPING, AND WHAT YOU SAY			
Phase	Time	Grouping	What happens
1. Blind redraw	5 min	Solo	Everything closed. Each student draws the mini visual on their prompt card from memory, on blank paper. No talking.
2. Swap and interrogate	5 min	Pairs	Trade papers. You mark your partner's sheet in the second colour. You may not explain, only ask. "Looks good" is not permitted.
3. Name the failure	3 min	Solo	Notes open now. For each gap your partner found, circle which of the three failures it was on the capture sheet.
4. Build the tool	7 min	Pairs	Pick your two worst pieces. Build a tool for each, then test it on your partner. Untested tools do not count.
5. Steal the best	3 min	Whole class	Each pair puts one tool on the board. Every student copies at least two they did not invent.

What to say

Opening, before phase 1

"Everything closed. Packets away, phones away, blank paper only. You have five minutes to draw the thing on your card from memory. If you cannot remember a piece, leave the space empty. An empty space is data. Filling it in with something vague is how you lose the information you need."

Setting up phase 2, this is the one that needs framing

"Trade papers. Your job is not to teach your partner. Your job is to find what is missing. Three questions only: what is missing, what is wrong, what is in the wrong order. Mark it in your second colour. If you hand a paper back and say it looks good, you have wasted your partner's five minutes."

Setting up phase 4

"A tool is a mnemonic, a drawing, an analogy, or a sentence you can say out loud. It has to attach the thing you cannot remember to something that already has structure. Then you test it: your partner covers the tool, and you produce the item from the tool alone. If you cannot, it is not finished."

Closing, after phase 5

"The empty spaces on your paper are the study list for this week. Not the whole chapter. Those spaces."

What to watch for while circulating

FAILURE MODES AND THE INTERVENTION		
What you see	What it means	What to do
A pair is silent in phase 2	They are being polite rather than useful	Ask one of them out loud: "name one thing missing on that page." Do not move on until they do.
A student redrew the diagram perfectly	They may have memorized the picture rather than the relationship	Cover it and ask them to explain it. If the explanation is thin, that is their piece.
A tool is just the list again	They restated instead of encoding	Ask what the tool attaches to. If the answer is nothing, it is not a tool.
Everyone circles "could not recall"	They are not distinguishing failure types	Read one application question aloud. Ask who knew the term but could not use it. That is the third failure.
A pair finishes phase 4 early	They picked easy pieces	Hand them a second card from the same set and restart at phase 1.

THE ONE RULE THAT MAKES OR BREAKS IT

Phase 1 has to be genuinely closed-book, and phase 2 has to be genuinely critical. If either softens, the activity turns into students showing each other their notes, which is the comfortable version and teaches nothing. The discomfort in the first ten minutes is the mechanism, not a side effect. Say that out loud to them the first time you run it, because otherwise the room will read your instructions as optional.

Adapting this to any module

Nothing in this activity is specific to Module 1 except the prompt cards. To run it on any other unit, replace Section 3 with cards naming that unit's mini visuals and its slippery pieces. Everything else, the phases, the timings, the script, the watch list, stays as written.

- **Card front**, one mini visual from that unit's pre-work packet. That is what gets redrawn.
- **Card back**, the two or three pieces students most often lose in that unit. Those seed the tool building in phase 4.
- One card per pair, and a spare set for the pairs who finish early.

SECTION 2

Student Capture Sheet

ONE PER STUDENT

Print one per student. They keep it. This is the only thing that leaves the room, and it is the study list for the week.

Repair Round

Name _____ Topic Date _____

1 What was on my card

2 What my partner found missing or wrong

3 How each one failed

THE PIECE

CIRCLE ONE

could not recall · recalled it wrong · knew it, could not apply

could not recall · recalled it wrong · knew it, could not apply

could not recall · recalled it wrong · knew it, could not apply

4 My two tools, tested on my partner

THE PIECE

MY TOOL

PARTNER TESTED IT

passed · not yet

passed · not yet

5 Two tools I stole from someone else

THIS WEEK'S STUDY LIST

The empty spaces on your redraw. Not the chapter, those spaces. Redraw the same visual on Thursday with everything closed and see which ones filled in.

CARD SET 1

Intro to Anatomy

CUT APART, ONE CARD PER PAIR

INTRO TO ANATOMY · CARD 1

REDRAW FROM MEMORY

Anatomical position with the three planes on it

IF YOU GET STUCK, THESE ARE THE PIECES MOST PEOPLE LOSE

- Proximal and distal, limbs only
- Midsagittal versus parasagittal

INTRO TO ANATOMY · CARD 2

REDRAW FROM MEMORY

The body cavities, hub and spokes

IF YOU GET STUCK, THESE ARE THE PIECES MOST PEOPLE LOSE

- Which membrane wraps which organ
- Borders and contents of the mediastinum

INTRO TO ANATOMY · CARD 3

REDRAW FROM MEMORY

Serous membrane, wall to organ as a chain

IF YOU GET STUCK, THESE ARE THE PIECES MOST PEOPLE LOSE

- Parietal versus visceral
- What the fluid is actually for

INTRO TO ANATOMY · CARD 4

REDRAW FROM MEMORY

Quadrants and the nine regions on one abdomen

IF YOU GET STUCK, THESE ARE THE PIECES MOST PEOPLE LOSE

- Nine regions versus four quadrants
- The retroperitoneal structures

CARD SET 2

Cell Anatomy

CUT APART, ONE CARD PER PAIR

CELL ANATOMY · CARD 1

REDRAW FROM MEMORY

The generalized cell, hub and spokes

IF YOU GET STUCK, THESE ARE THE PIECES MOST PEOPLE LOSE

- Cytoplasm versus cytosol
- Free versus bound ribosomes

CELL ANATOMY · CARD 2

REDRAW FROM MEMORY

The secretory pathway as a chain

IF YOU GET STUCK, THESE ARE THE PIECES MOST PEOPLE LOSE

- The order of the pathway
- Where the protein is modified versus made

CELL ANATOMY · CARD 3

REDRAW FROM MEMORY

Rough ER against smooth ER as a grid

IF YOU GET STUCK, THESE ARE THE PIECES MOST PEOPLE LOSE

- Rough versus smooth ER
- Lysosome versus peroxisome

CELL ANATOMY · CARD 4

REDRAW FROM MEMORY

9 plus 0 against 9 plus 2 as a split

IF YOU GET STUCK, THESE ARE THE PIECES MOST PEOPLE LOSE

- Centrioles versus cilia
- Which arrangement moves

CARD SET 3

Tissues and Histology

CUT APART, ONE CARD PER PAIR

TISSUES AND HISTOLOGY · CARD 1

REDRAW FROM MEMORY

Epithelium, layers crossed with shape

IF YOU GET STUCK, THESE ARE THE PIECES MOST PEOPLE LOSE

- Simple versus stratified, and which layer you name
- Pseudostratified, why it only looks layered

TISSUES AND HISTOLOGY · CARD 2

REDRAW FROM MEMORY

The five junctions on two adjacent cells

IF YOU GET STUCK, THESE ARE THE PIECES MOST PEOPLE LOSE

- Desmosome versus hemidesmosome
- Which junction seals and which communicates

TISSUES AND HISTOLOGY · CARD 3

REDRAW FROM MEMORY

Connective tissue, what it is made of

IF YOU GET STUCK, THESE ARE THE PIECES MOST PEOPLE LOSE

- Collagen versus reticular versus elastic
- Which connective tissues are avascular

TISSUES AND HISTOLOGY · CARD 4

REDRAW FROM MEMORY

The three gland modes as three chains

IF YOU GET STUCK, THESE ARE THE PIECES MOST PEOPLE LOSE

- Merocrine versus apocrine versus holocrine
- Blasts versus cytes

CARD SET 4

Integumentary System

CUT APART, ONE CARD PER PAIR

INTEGUMENTARY SYSTEM · CARD 1

REDRAW FROM MEMORY

The five strata as a ladder

IF YOU GET STUCK, THESE ARE THE PIECES MOST PEOPLE LOSE

- The order of the strata
- Which layer is missing in thin skin

INTEGUMENTARY SYSTEM · CARD 2

REDRAW FROM MEMORY

Melanin transfer as a chain

IF YOU GET STUCK, THESE ARE THE PIECES MOST PEOPLE LOSE

- Why the granules sit over the nucleus
- Eumelanin versus pheomelanin

INTEGUMENTARY SYSTEM · CARD 3

REDRAW FROM MEMORY

The four skin glands as a grid

IF YOU GET STUCK, THESE ARE THE PIECES MOST PEOPLE LOSE

- Eccrine versus apocrine
- Which gland opens onto a follicle

INTEGUMENTARY SYSTEM · CARD 4

REDRAW FROM MEMORY

The dermis, papillary against reticular

IF YOU GET STUCK, THESE ARE THE PIECES MOST PEOPLE LOSE

- Papillary versus reticular
- The three structures in a dermal papilla